

--DROP TABLES

```
DROP TABLE emppayroll CASCADE CONSTRAINTS;  
DROP TABLE payments CASCADE CONSTRAINTS;  
DROP TABLE tickets CASCADE CONSTRAINTS;  
DROP TABLE shifts CASCADE CONSTRAINTS;  
DROP TABLE orders CASCADE CONSTRAINTS;  
DROP TABLE events CASCADE CONSTRAINTS;  
DROP TABLE artists CASCADE CONSTRAINTS;  
DROP TABLE customers CASCADE CONSTRAINTS;  
DROP TABLE employees CASCADE CONSTRAINTS;  
DROP TABLE venue CASCADE CONSTRAINTS;
```

– CREATE TABLES

-- 1. Artists Table

```
CREATE TABLE artists (  
    artistid NUMBER(8) NOT NULL,  
    name    VARCHAR2(100) NOT NULL,  
    genre   VARCHAR2(150) NOT NULL,  
    CONSTRAINT artists_pk PRIMARY KEY (artistid)  
);
```

-- 2. Venue Table

```
CREATE TABLE venue (  
    venueid NUMBER(8) NOT NULL,  
    city    VARCHAR2(150) NOT NULL,  
    state   VARCHAR2(50),  
    type    VARCHAR2(100) NOT NULL,  
    CONSTRAINT venue_pk PRIMARY KEY (venueid)  
);
```

-- 3. Customers Table

```
CREATE TABLE customers (  
    customerid NUMBER(8) NOT NULL,  
    name        VARCHAR2(100),  
    type        VARCHAR2(100),  
    email       VARCHAR2(100) NOT NULL,  
    phone       VARCHAR2(20),  
    address     VARCHAR2(150),  
    CONSTRAINT customers_pk PRIMARY KEY (customerid)  
);
```

-- 4. Employees Table

```
CREATE TABLE employees (  
    empid NUMBER(8) NOT NULL,  
    name VARCHAR2(100) NOT NULL,  
    dept VARCHAR2(100),  
    salary NUMBER(10, 2) NOT NULL,  
    CONSTRAINT employees_pk PRIMARY KEY (empid)  
);
```

-- 5. EmpPayroll Table

```
CREATE TABLE emppayroll (  
    emppayrollid NUMBER(8) NOT NULL,  
    empid NUMBER(8) NOT NULL,  
    hoursworked NUMBER(8, 2) NOT NULL,  
    hourlyrate NUMBER(8, 2) NOT NULL,  
    amtpaid NUMBER(10, 2) NOT NULL,  
    CONSTRAINT emppayroll_pk PRIMARY KEY (emppayrollid)  
);
```

-- 6. Events Table

```
CREATE TABLE events (  
    eventid NUMBER(8) NOT NULL,  
    type VARCHAR2(100),  
    event_date DATE NOT NULL,  
    description VARCHAR2(1000),  
    artistid NUMBER(8) NOT NULL,  
    venueid NUMBER(8) NOT NULL,  
    CONSTRAINT events_pk PRIMARY KEY (eventid)  
);
```

-- 7. Orders Table

```
CREATE TABLE orders (  
    orderid NUMBER(8) NOT NULL,  
    order_date DATE,  
    shipdate DATE,  
    eventid NUMBER(8) NOT NULL,  
    customerid NUMBER(8) NOT NULL,  
    CONSTRAINT orders_pk PRIMARY KEY (orderid)  
);
```

-- 8. Payments Table

```
CREATE TABLE payments (  
    paymentid NUMBER(8) NOT NULL,  
    orderid NUMBER(8) NOT NULL,  
    pmtamount NUMBER (10,2) NOT NULL,
```

```
CONSTRAINT payments_pk PRIMARY KEY (paymentid)
);
```

-- 9. Shifts Table

```
CREATE TABLE shifts (
    empid  NUMBER(8) NOT NULL,
    eventid NUMBER(8) NOT NULL,
    duration NUMBER (8,2),
    CONSTRAINT shifts_pk PRIMARY KEY (empid, eventid)
);
```

-- 10. Tickets Table

```
CREATE TABLE tickets (
    ticketid NUMBER(8) NOT NULL,
    orderid  NUMBER(8),
    type     VARCHAR2(100),
    eventid  NUMBER(8) NOT NULL,
    CONSTRAINT tickets_pk PRIMARY KEY (ticketid)
);
```

ALTER TABLE emppayroll

```
ADD CONSTRAINT emppayroll_employees_fk FOREIGN KEY (empid)
REFERENCES employees (empid);
```

ALTER TABLE events

```
ADD CONSTRAINT events_artists_fk FOREIGN KEY (artistid)
REFERENCES artists (artistid);
```

ALTER TABLE events

```
ADD CONSTRAINT events_venue_fk FOREIGN KEY (venueid)
REFERENCES venue (venueid);
```

ALTER TABLE orders

```
ADD CONSTRAINT orders_customers_fk FOREIGN KEY (customerid)
REFERENCES customers (customerid);
```

ALTER TABLE orders

```
ADD CONSTRAINT orders_events_fk FOREIGN KEY (eventid)
REFERENCES events (eventid);
```

ALTER TABLE payments

```
ADD CONSTRAINT payments_orders_fk FOREIGN KEY (orderid)
REFERENCES orders (orderid);
```

```
ALTER TABLE shifts
  ADD CONSTRAINT shifts_employees_fk FOREIGN KEY (empid)
    REFERENCES employees (empid);
```

```
ALTER TABLE shifts
  ADD CONSTRAINT shifts_events_fk FOREIGN KEY (eventid)
    REFERENCES events (eventid);
```

```
ALTER TABLE tickets
  ADD CONSTRAINT tickets_events_fk FOREIGN KEY (eventid)
    REFERENCES events (eventid);
```

```
ALTER TABLE tickets
  ADD CONSTRAINT tickets_orders_fk FOREIGN KEY (orderid)
    REFERENCES orders (orderid);
```

```
CREATE OR REPLACE TRIGGER trg_assign_ticket
AFTER INSERT ON orders
FOR EACH ROW
BEGIN
  -- Update a single available ticket for the requested event
  UPDATE tickets
  SET orderid = :NEW.orderid
  WHERE ticketid = (
    -- Subquery to find the first available ticket ID for this event
    SELECT MIN(ticketid)
    FROM tickets
    WHERE eventid = :NEW.eventid
    AND orderid IS NULL
  );
END;
/
```

```
-- =====
-- 1. POPULATE ARTISTS (15 Records)
-- =====
```

```
INSERT INTO artists (artistid, name, genre) VALUES (1, 'The Midnight Echo', 'Indie Rock');
INSERT INTO artists (artistid, name, genre) VALUES (2, 'Synthwave Dreams', 'Electronic');
INSERT INTO artists (artistid, name, genre) VALUES (3, 'Sarah Jenkins', 'Folk');
INSERT INTO artists (artistid, name, genre) VALUES (4, 'Velvet Underground Cover', 'Rock');
INSERT INTO artists (artistid, name, genre) VALUES (5, 'DJ Quantum', 'EDM');
INSERT INTO artists (artistid, name, genre) VALUES (6, 'Blue Note Quartet', 'Jazz');
INSERT INTO artists (artistid, name, genre) VALUES (7, 'Steel Panther', 'Heavy Metal');
INSERT INTO artists (artistid, name, genre) VALUES (8, 'Luna and The Whalers', 'Pop');
```

```

INSERT INTO artists (artistid, name, genre) VALUES (9, 'Cactus Jack', 'Country');
INSERT INTO artists (artistid, name, genre) VALUES (10, 'Neon Skyline', 'Synth-Pop');
INSERT INTO artists (artistid, name, genre) VALUES (11, 'The Crimson Tide', 'Punk');
INSERT INTO artists (artistid, name, genre) VALUES (12, 'Orchestral Maneuvers', 'Classical');
INSERT INTO artists (artistid, name, genre) VALUES (13, 'River City Beats', 'Hip Hop');
INSERT INTO artists (artistid, name, genre) VALUES (14, 'Misty Mountains', 'Bluegrass');
INSERT INTO artists (artistid, name, genre) VALUES (15, 'Solar Flare', 'Alternative');

```

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-- =====
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```
-- 2. POPULATE VENUE (10 Records)
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-- =====
```

```

INSERT INTO venue (venueid, city, state, type) VALUES (101, 'Austin', 'TX', 'Amphitheater');
INSERT INTO venue (venueid, city, state, type) VALUES (102, 'Nashville', 'TN', 'Club');
INSERT INTO venue (venueid, city, state, type) VALUES (103, 'New York', 'NY', 'Arena');
INSERT INTO venue (venueid, city, state, type) VALUES (104, 'Chicago', 'IL', 'Theater');
INSERT INTO venue (venueid, city, state, type) VALUES (105, 'Seattle', 'WA', 'Outdoors');
INSERT INTO venue (venueid, city, state, type) VALUES (106, 'Denver', 'CO', 'Ballroom');
INSERT INTO venue (venueid, city, state, type) VALUES (107, 'Miami', 'FL', 'Beach Club');
INSERT INTO venue (venueid, city, state, type) VALUES (108, 'San Francisco', 'CA', 'Hall');
INSERT INTO venue (venueid, city, state, type) VALUES (109, 'Atlanta', 'GA', 'Stadium');
INSERT INTO venue (venueid, city, state, type) VALUES (110, 'Boston', 'MA', 'Opera House');

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```

```
-- 3. POPULATE CUSTOMERS (30 Records)
```

```
-- =====
```

```

INSERT INTO customers (customerid, name, type, email, phone, address) VALUES (201, 'Alice Thompson', 'VIP', 'alice.t@gmail.com', '555-0101', '123 Maple St');
INSERT INTO customers (customerid, name, type, email, phone, address) VALUES (202, 'Bob Richards', 'Regular', 'bob.r@yahoo.com', '555-0102', '456 Oak Ave');
INSERT INTO customers (customerid, name, type, email, phone, address) VALUES (203, 'Charlie Davis', 'Regular', 'charlie.d@outlook.com', '555-0103', '789 Pine Rd');
INSERT INTO customers (customerid, name, type, email, phone, address) VALUES (204, 'Diana Prince', 'VIP', 'diana.p@gmail.com', '555-0104', '101 Amazon Way');
INSERT INTO customers (customerid, name, type, email, phone, address) VALUES (205, 'Edward Norton', 'Student', 'ed.n@edu.com', '555-0105', '202 College Ln');
INSERT INTO customers (customerid, name, type, email, phone, address) VALUES (206, 'Fiona Glenanne', 'Regular', 'fiona.g@cia.gov', '555-0106', '303 Burn St');
INSERT INTO customers (customerid, name, type, email, phone, address) VALUES (207, 'George Costanza', 'Regular', 'george.c@vandelay.com', '555-0107', '404 Queens Blvd');
INSERT INTO customers (customerid, name, type, email, phone, address) VALUES (208, 'Hannah Abbott', 'Student', 'hannah.a@hogwarts.uk', '555-0108', '505 Hufflepuff Rd');
INSERT INTO customers (customerid, name, type, email, phone, address) VALUES (209, 'Ian Wright', 'Regular', 'ian.w@arsenal.com', '555-0109', '606 Highbury Ave');

```

```

INSERT INTO customers (customerid, name, type, email, phone, address) VALUES (210, 'Julia
Roberts', 'VIP', 'julia.r@hollywood.com', '555-0110', '707 Rodeo Dr');
INSERT INTO customers (customerid, name, type, email, phone, address) VALUES (211, 'Kevin
Hart', 'Regular', 'kevin.h@comedy.com', '555-0111', '808 Laugh Ln');
INSERT INTO customers (customerid, name, type, email, phone, address) VALUES (212,
'Laura Palmer', 'Regular', 'laura.p@twinpeaks.com', '555-0112', '705 Pine St');
INSERT INTO customers (customerid, name, type, email, phone, address) VALUES (213,
'Michael Scott', 'Regular', 'm.scott@dundermifflin.com', '555-0113', '1725 Slough Ave');
INSERT INTO customers (customerid, name, type, email, phone, address) VALUES (214,
'Nancy Drew', 'Student', 'n.drew@mysteries.com', '555-0114', '221 River St');
INSERT INTO customers (customerid, name, type, email, phone, address) VALUES (215,
'Oscar Martinez', 'Regular', 'o.martinez@accounting.com', '555-0115', '120 Paper Way');
INSERT INTO customers (customerid, name, type, email, phone, address) VALUES (216, 'Peter
Parker', 'Student', 'p.parker@dailybugle.com', '555-0116', '20 Ingram St');
INSERT INTO customers (customerid, name, type, email, phone, address) VALUES (217,
'Quentin Smith', 'Regular', 'q.smith@dreamer.com', '555-0117', '1428 Elm St');
INSERT INTO customers (customerid, name, type, email, phone, address) VALUES (218,
'Rachel Green', 'Regular', 'r.green@centralperk.com', '555-0118', '90 Bedford St');
INSERT INTO customers (customerid, name, type, email, phone, address) VALUES (219,
'Steve Rogers', 'VIP', 's.rogers@avengers.org', '555-0119', '569 Lefferts Ave');
INSERT INTO customers (customerid, name, type, email, phone, address) VALUES (220, 'Tony
Stark', 'VIP', 't.stark@starkintl.com', '555-0120', '10880 Malibu Rd');
INSERT INTO customers (customerid, name, type, email, phone, address) VALUES (221,
'Ursula Buffay', 'Regular', 'u.buffay@waitress.com', '555-0121', '455 West 11th St');
INSERT INTO customers (customerid, name, type, email, phone, address) VALUES (222,
'Victor Von Doom', 'VIP', 'v.doom@latveria.gov', '555-0122', '1 Castle Dr');
INSERT INTO customers (customerid, name, type, email, phone, address) VALUES (223,
'Wanda Maximoff', 'Regular', 'w.maximoff@chaos.com', '555-0123', '2800 Sherwood Dr');
INSERT INTO customers (customerid, name, type, email, phone, address) VALUES (224,
'Xavier Charles', 'VIP', 'professor.x@mansion.edu', '555-0124', '1407 Graymalkin Ln');
INSERT INTO customers (customerid, name, type, email, phone, address) VALUES (225,
'Yvonne Strahovski', 'Regular', 'yvonne.s@agent.com', '555-0125', '999 Secret Ave');
INSERT INTO customers (customerid, name, type, email, phone, address) VALUES (226, 'Zane
Grey', 'Regular', 'z.grey@west.com', '555-0126', '12 Cowboy Way');
INSERT INTO customers (customerid, name, type, email, phone, address) VALUES (227,
'Aaron Burr', 'VIP', 'a.burr@hamilton.com', '555-0127', '18 Maiden Ln');
INSERT INTO customers (customerid, name, type, email, phone, address) VALUES (228, 'Betty
Cooper', 'Student', 'b.cooper@riverdale.com', '555-0128', '123 Elm St');
INSERT INTO customers (customerid, name, type, email, phone, address) VALUES (229, 'Clark
Kent', 'Regular', 'c.kent@dailyplanet.com', '555-0129', '344 Clinton St');
INSERT INTO customers (customerid, name, type, email, phone, address) VALUES (230, 'Zoe
Saldana', 'VIP', 'zoe.s@space.com', '555-0130', '909 Star Way');

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```
-- 4. POPULATE EMPLOYEES (12 Records)
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```

-- =====
INSERT INTO employees (empid, name, dept, salary) VALUES (501, 'John Smith',
'Management', 75000.00);
INSERT INTO employees (empid, name, dept, salary) VALUES (502, 'Jane Doe', 'Operations',
55000.00);
INSERT INTO employees (empid, name, dept, salary) VALUES (503, 'Mike Miller', 'Security',
45000.00);
INSERT INTO employees (empid, name, dept, salary) VALUES (504, 'Sarah Wilson', 'Ticketing',
42000.00);
INSERT INTO employees (empid, name, dept, salary) VALUES (505, 'Tom Brown', 'Operations',
52000.00);
INSERT INTO employees (empid, name, dept, salary) VALUES (506, 'Emily Davis', 'Marketing',
60000.00);
INSERT INTO employees (empid, name, dept, salary) VALUES (507, 'Chris Evans', 'Security',
46000.00);
INSERT INTO employees (empid, name, dept, salary) VALUES (508, 'Anna Kendrick',
'Ticketing', 41000.00);
INSERT INTO employees (empid, name, dept, salary) VALUES (509, 'Robert Downey',
'Management', 90000.00);
INSERT INTO employees (empid, name, dept, salary) VALUES (510, 'Scarlett J', 'Marketing',
62000.00);
INSERT INTO employees (empid, name, dept, salary) VALUES (511, 'Mark Ruffalo',
'Operations', 54000.00);
INSERT INTO employees (empid, name, dept, salary) VALUES (512, 'Jeremy Renner',
'Security', 47000.00);

```

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-- =====
-- 5. POPULATE EMPPAYROLL (12 Records)
-- =====

```

```

INSERT INTO emppayroll (emppayrollid, empid, hoursworked, hourlyrate, amtpaid) VALUES (1,
501, 40, 36.00, 1440.00);
INSERT INTO emppayroll (emppayrollid, empid, hoursworked, hourlyrate, amtpaid) VALUES (2,
502, 40, 26.50, 1060.00);
INSERT INTO emppayroll (emppayrollid, empid, hoursworked, hourlyrate, amtpaid) VALUES (3,
503, 45, 20.00, 950.00);
INSERT INTO emppayroll (emppayrollid, empid, hoursworked, hourlyrate, amtpaid) VALUES (4,
504, 38, 20.00, 760.00);
INSERT INTO emppayroll (emppayrollid, empid, hoursworked, hourlyrate, amtpaid) VALUES (5,
505, 40, 25.00, 1000.00);
INSERT INTO emppayroll (emppayrollid, empid, hoursworked, hourlyrate, amtpaid) VALUES (6,
506, 40, 28.80, 1152.00);
INSERT INTO emppayroll (emppayrollid, empid, hoursworked, hourlyrate, amtpaid) VALUES (7,
507, 50, 21.00, 1155.00);

```

```

INSERT INTO emppayroll (emppayrollid, empid, hoursworked, hourlyrate, amtpaid) VALUES (8,
508, 30, 20.00, 600.00);
INSERT INTO emppayroll (emppayrollid, empid, hoursworked, hourlyrate, amtpaid) VALUES (9,
509, 40, 43.00, 1720.00);
INSERT INTO emppayroll (emppayrollid, empid, hoursworked, hourlyrate, amtpaid) VALUES
(10, 510, 40, 29.80, 1192.00);
INSERT INTO emppayroll (emppayrollid, empid, hoursworked, hourlyrate, amtpaid) VALUES
(11, 511, 40, 26.00, 1040.00);
INSERT INTO emppayroll (emppayrollid, empid, hoursworked, hourlyrate, amtpaid) VALUES
(12, 512, 42, 22.50, 990.00);

```

```
-- =====
```

```
-- 6. POPULATE EVENTS (10 Records)
```

```
-- =====
```

```

INSERT INTO events (eventid, type, event_date, description, artistid, venueid) VALUES (301,
'Concert', TO_DATE('2026-05-15','YYYY-MM-DD'), 'Summer Kickoff', 1, 101);
INSERT INTO events (eventid, type, event_date, description, artistid, venueid) VALUES (302,
'Live Recording', TO_DATE('2026-05-20','YYYY-MM-DD'), 'Live at Nashville', 2, 102);
INSERT INTO events (eventid, type, event_date, description, artistid, venueid) VALUES (303,
'Festival', TO_DATE('2026-06-01','YYYY-MM-DD'), 'The Big Apple Fest', 3, 103);
INSERT INTO events (eventid, type, event_date, description, artistid, venueid) VALUES (304,
'Charity', TO_DATE('2026-06-10','YYYY-MM-DD'), 'Acoustic Night', 4, 104);
INSERT INTO events (eventid, type, event_date, description, artistid, venueid) VALUES (305,
'Concert', TO_DATE('2026-06-15','YYYY-MM-DD'), 'Neon Nights', 5, 105);
INSERT INTO events (eventid, type, event_date, description, artistid, venueid) VALUES (306,
'Gala', TO_DATE('2026-07-04','YYYY-MM-DD'), 'Independence Jazz', 6, 106);
INSERT INTO events (eventid, type, event_date, description, artistid, venueid) VALUES (307,
'Concert', TO_DATE('2026-07-12','YYYY-MM-DD'), 'Metal Mania', 7, 107);
INSERT INTO events (eventid, type, event_date, description, artistid, venueid) VALUES (308,
'Concert', TO_DATE('2026-07-20','YYYY-MM-DD'), 'Pop Princess Tour', 8, 108);
INSERT INTO events (eventid, type, event_date, description, artistid, venueid) VALUES (309,
'Festival', TO_DATE('2026-08-05','YYYY-MM-DD'), 'Country Roots', 9, 109);
INSERT INTO events (eventid, type, event_date, description, artistid, venueid) VALUES (310,
'Opera', TO_DATE('2026-08-15','YYYY-MM-DD'), 'Modern Synth Opera', 10, 110);

```

```
-- =====
```

```
-- 7. POPULATE ORDERS (25 Records)
```

```
-- =====
```

```

INSERT INTO orders (orderid, order_date, shipdate, eventid, customerid) VALUES (401,
TO_DATE('2026-04-01','YYYY-MM-DD'), TO_DATE('2026-04-05','YYYY-MM-DD'), 301, 201);
INSERT INTO orders (orderid, order_date, shipdate, eventid, customerid) VALUES (402,
TO_DATE('2026-04-02','YYYY-MM-DD'), TO_DATE('2026-04-06','YYYY-MM-DD'), 301, 202);
INSERT INTO orders (orderid, order_date, shipdate, eventid, customerid) VALUES (403,
TO_DATE('2026-04-05','YYYY-MM-DD'), TO_DATE('2026-04-10','YYYY-MM-DD'), 302, 203);

```



```

INSERT INTO orders (orderid, order_date, shipdate, eventid, customerid) VALUES (404,
TO_DATE('2026-04-10','YYYY-MM-DD'), NULL, 303, 204);
INSERT INTO orders (orderid, order_date, shipdate, eventid, customerid) VALUES (405,
TO_DATE('2026-04-12','YYYY-MM-DD'), TO_DATE('2026-04-15','YYYY-MM-DD'), 303, 205);
INSERT INTO orders (orderid, order_date, shipdate, eventid, customerid) VALUES (406,
TO_DATE('2026-04-15','YYYY-MM-DD'), TO_DATE('2026-04-20','YYYY-MM-DD'), 304, 206);
INSERT INTO orders (orderid, order_date, shipdate, eventid, customerid) VALUES (407,
TO_DATE('2026-04-18','YYYY-MM-DD'), TO_DATE('2026-04-22','YYYY-MM-DD'), 304, 207);
INSERT INTO orders (orderid, order_date, shipdate, eventid, customerid) VALUES (408,
TO_DATE('2026-04-20','YYYY-MM-DD'), NULL, 305, 208);
INSERT INTO orders (orderid, order_date, shipdate, eventid, customerid) VALUES (409,
TO_DATE('2026-04-22','YYYY-MM-DD'), TO_DATE('2026-04-28','YYYY-MM-DD'), 305, 209);
INSERT INTO orders (orderid, order_date, shipdate, eventid, customerid) VALUES (410,
TO_DATE('2026-04-25','YYYY-MM-DD'), TO_DATE('2026-04-30','YYYY-MM-DD'), 306, 210);
INSERT INTO orders (orderid, order_date, shipdate, eventid, customerid) VALUES (411,
TO_DATE('2026-04-28','YYYY-MM-DD'), TO_DATE('2026-05-02','YYYY-MM-DD'), 306, 211);
INSERT INTO orders (orderid, order_date, shipdate, eventid, customerid) VALUES (412,
TO_DATE('2026-05-01','YYYY-MM-DD'), NULL, 307, 212);
INSERT INTO orders (orderid, order_date, shipdate, eventid, customerid) VALUES (413,
TO_DATE('2026-05-03','YYYY-MM-DD'), TO_DATE('2026-05-08','YYYY-MM-DD'), 307, 213);
INSERT INTO orders (orderid, order_date, shipdate, eventid, customerid) VALUES (414,
TO_DATE('2026-05-05','YYYY-MM-DD'), TO_DATE('2026-05-10','YYYY-MM-DD'), 308, 214);
INSERT INTO orders (orderid, order_date, shipdate, eventid, customerid) VALUES (415,
TO_DATE('2026-05-07','YYYY-MM-DD'), TO_DATE('2026-05-12','YYYY-MM-DD'), 308, 215);
INSERT INTO orders (orderid, order_date, shipdate, eventid, customerid) VALUES (416,
TO_DATE('2026-05-10','YYYY-MM-DD'), NULL, 309, 216);
INSERT INTO orders (orderid, order_date, shipdate, eventid, customerid) VALUES (417,
TO_DATE('2026-05-12','YYYY-MM-DD'), TO_DATE('2026-05-17','YYYY-MM-DD'), 309, 217);
INSERT INTO orders (orderid, order_date, shipdate, eventid, customerid) VALUES (418,
TO_DATE('2026-05-15','YYYY-MM-DD'), TO_DATE('2026-05-20','YYYY-MM-DD'), 310, 218);
INSERT INTO orders (orderid, order_date, shipdate, eventid, customerid) VALUES (419,
TO_DATE('2026-05-18','YYYY-MM-DD'), TO_DATE('2026-05-23','YYYY-MM-DD'), 301, 219);
INSERT INTO orders (orderid, order_date, shipdate, eventid, customerid) VALUES (420,
TO_DATE('2026-05-20','YYYY-MM-DD'), NULL, 302, 220);
INSERT INTO orders (orderid, order_date, shipdate, eventid, customerid) VALUES (421,
TO_DATE('2026-05-22','YYYY-MM-DD'), TO_DATE('2026-05-27','YYYY-MM-DD'), 303, 221);
INSERT INTO orders (orderid, order_date, shipdate, eventid, customerid) VALUES (422,
TO_DATE('2026-05-25','YYYY-MM-DD'), TO_DATE('2026-05-30','YYYY-MM-DD'), 304, 222);
INSERT INTO orders (orderid, order_date, shipdate, eventid, customerid) VALUES (423,
TO_DATE('2026-05-28','YYYY-MM-DD'), TO_DATE('2026-06-02','YYYY-MM-DD'), 305, 223);
INSERT INTO orders (orderid, order_date, shipdate, eventid, customerid) VALUES (424,
TO_DATE('2026-05-30','YYYY-MM-DD'), NULL, 306, 224);
INSERT INTO orders (orderid, order_date, shipdate, eventid, customerid) VALUES (425,
TO_DATE('2026-06-01','YYYY-MM-DD'), TO_DATE('2026-06-05','YYYY-MM-DD'), 310, 230);

```

-- =====

-- 8. POPULATE TICKETS (50 Records)

-- =====

-- Tickets assigned to Orders (first 40)

```
INSERT INTO tickets (ticketid, orderid, type, eventid) VALUES (1001, 401, 'VIP', 301);
INSERT INTO tickets (ticketid, orderid, type, eventid) VALUES (1002, 401, 'VIP', 301);
INSERT INTO tickets (ticketid, orderid, type, eventid) VALUES (1003, 402, 'GA', 301);
INSERT INTO tickets (ticketid, orderid, type, eventid) VALUES (1004, 403, 'GA', 302);
INSERT INTO tickets (ticketid, orderid, type, eventid) VALUES (1005, 404, 'Premium', 303);
INSERT INTO tickets (ticketid, orderid, type, eventid) VALUES (1006, 405, 'GA', 303);
INSERT INTO tickets (ticketid, orderid, type, eventid) VALUES (1007, 406, 'GA', 304);
INSERT INTO tickets (ticketid, orderid, type, eventid) VALUES (1008, 407, 'GA', 304);
INSERT INTO tickets (ticketid, orderid, type, eventid) VALUES (1009, 408, 'VIP', 305);
INSERT INTO tickets (ticketid, orderid, type, eventid) VALUES (1010, 409, 'GA', 305);
INSERT INTO tickets (ticketid, orderid, type, eventid) VALUES (1011, 410, 'VIP', 306);
INSERT INTO tickets (ticketid, orderid, type, eventid) VALUES (1012, 411, 'GA', 306);
INSERT INTO tickets (ticketid, orderid, type, eventid) VALUES (1013, 412, 'GA', 307);
INSERT INTO tickets (ticketid, orderid, type, eventid) VALUES (1014, 413, 'GA', 307);
INSERT INTO tickets (ticketid, orderid, type, eventid) VALUES (1015, 414, 'GA', 308);
INSERT INTO tickets (ticketid, orderid, type, eventid) VALUES (1016, 415, 'GA', 308);
INSERT INTO tickets (ticketid, orderid, type, eventid) VALUES (1017, 416, 'VIP', 309);
INSERT INTO tickets (ticketid, orderid, type, eventid) VALUES (1018, 417, 'GA', 309);
INSERT INTO tickets (ticketid, orderid, type, eventid) VALUES (1019, 418, 'Premium', 310);
INSERT INTO tickets (ticketid, orderid, type, eventid) VALUES (1020, 419, 'VIP', 301);
INSERT INTO tickets (ticketid, orderid, type, eventid) VALUES (1021, 420, 'GA', 302);
INSERT INTO tickets (ticketid, orderid, type, eventid) VALUES (1022, 421, 'GA', 303);
INSERT INTO tickets (ticketid, orderid, type, eventid) VALUES (1023, 422, 'GA', 304);
INSERT INTO tickets (ticketid, orderid, type, eventid) VALUES (1024, 423, 'GA', 305);
INSERT INTO tickets (ticketid, orderid, type, eventid) VALUES (1025, 424, 'GA', 306);
INSERT INTO tickets (ticketid, orderid, type, eventid) VALUES (1026, 425, 'Premium', 310);
INSERT INTO tickets (ticketid, orderid, type, eventid) VALUES (1027, 401, 'GA', 301);
INSERT INTO tickets (ticketid, orderid, type, eventid) VALUES (1028, 402, 'GA', 301);
INSERT INTO tickets (ticketid, orderid, type, eventid) VALUES (1029, 403, 'GA', 302);
INSERT INTO tickets (ticketid, orderid, type, eventid) VALUES (1030, 404, 'GA', 303);
INSERT INTO tickets (ticketid, orderid, type, eventid) VALUES (1031, 405, 'GA', 303);
INSERT INTO tickets (ticketid, orderid, type, eventid) VALUES (1032, 406, 'GA', 304);
INSERT INTO tickets (ticketid, orderid, type, eventid) VALUES (1033, 407, 'GA', 304);
INSERT INTO tickets (ticketid, orderid, type, eventid) VALUES (1034, 408, 'GA', 305);
INSERT INTO tickets (ticketid, orderid, type, eventid) VALUES (1035, 409, 'GA', 305);
INSERT INTO tickets (ticketid, orderid, type, eventid) VALUES (1036, 410, 'GA', 306);
INSERT INTO tickets (ticketid, orderid, type, eventid) VALUES (1037, 411, 'GA', 306);
INSERT INTO tickets (ticketid, orderid, type, eventid) VALUES (1038, 412, 'GA', 307);
INSERT INTO tickets (ticketid, orderid, type, eventid) VALUES (1039, 413, 'GA', 307);
```

```

INSERT INTO tickets (ticketid, orderid, type, eventid) VALUES (1040, 414, 'GA', 308);

-- Unsold Inventory (Order ID is NULL)
INSERT INTO tickets (ticketid, orderid, type, eventid) VALUES (1041, NULL, 'GA', 301);
INSERT INTO tickets (ticketid, orderid, type, eventid) VALUES (1042, NULL, 'VIP', 302);
INSERT INTO tickets (ticketid, orderid, type, eventid) VALUES (1043, NULL, 'GA', 303);
INSERT INTO tickets (ticketid, orderid, type, eventid) VALUES (1044, NULL, 'Premium', 304);
INSERT INTO tickets (ticketid, orderid, type, eventid) VALUES (1045, NULL, 'GA', 301);
INSERT INTO tickets (ticketid, orderid, type, eventid) VALUES (1046, NULL, 'VIP', 302);
INSERT INTO tickets (ticketid, orderid, type, eventid) VALUES (1047, NULL, 'GA', 309);
INSERT INTO tickets (ticketid, orderid, type, eventid) VALUES (1048, NULL, 'GA', 310);
INSERT INTO tickets (ticketid, orderid, type, eventid) VALUES (1049, NULL, 'Premium', 310);
INSERT INTO tickets (ticketid, orderid, type, eventid) VALUES (1050, NULL, 'GA', 310);

-- =====
-- 9. POPULATE PAYMENTS (25 Records)
-- =====
-- One payment for each order
INSERT INTO payments (paymentid, orderid) VALUES (601, 401);
INSERT INTO payments (paymentid, orderid) VALUES (602, 402);
INSERT INTO payments (paymentid, orderid) VALUES (603, 403);
INSERT INTO payments (paymentid, orderid) VALUES (604, 404);
INSERT INTO payments (paymentid, orderid) VALUES (605, 405);
-- ... [One-to-one mapping for all 25 orders]
INSERT INTO payments (paymentid, orderid) VALUES (625, 425);

-- =====
-- 10. POPULATE SHIFTS (20 Records)
-- =====
-- Assigning employees to multiple events
INSERT INTO shifts (empid, eventid) VALUES (502, 301);
INSERT INTO shifts (empid, eventid) VALUES (503, 301);
INSERT INTO shifts (empid, eventid) VALUES (507, 301);
INSERT INTO shifts (empid, eventid) VALUES (512, 301);
INSERT INTO shifts (empid, eventid) VALUES (502, 302);
INSERT INTO shifts (empid, eventid) VALUES (503, 302);
INSERT INTO shifts (empid, eventid) VALUES (504, 303);
INSERT INTO shifts (empid, eventid) VALUES (505, 304);
INSERT INTO shifts (empid, eventid) VALUES (506, 305);
INSERT INTO shifts (empid, eventid) VALUES (507, 306);
INSERT INTO shifts (empid, eventid) VALUES (511, 307);
INSERT INTO shifts (empid, eventid) VALUES (512, 308);
INSERT INTO shifts (empid, eventid) VALUES (501, 309);
INSERT INTO shifts (empid, eventid) VALUES (509, 310);

```

```
-- Adding double shifts for some
INSERT INTO shifts (empid, eventid) VALUES (502, 303);
INSERT INTO shifts (empid, eventid) VALUES (503, 303);
INSERT INTO shifts (empid, eventid) VALUES (511, 301);
INSERT INTO shifts (empid, eventid) VALUES (504, 305);
INSERT INTO shifts (empid, eventid) VALUES (508, 301);
INSERT INTO shifts (empid, eventid) VALUES (508, 302);
```

```
COMMIT;
```

```
--Queries
```

```
SELECT name, dept, salary
FROM employees
WHERE salary > (SELECT AVG(salary) FROM employees);
```

```
SELECT e.event_date, e.description, a.name AS artist_name, v.city, v.type AS venue_type
FROM events e
JOIN artists a ON e.artistid = a.artistid
JOIN venue v ON e.venueid = v.venueid
ORDER BY e.event_date;
```

```
SELECT t.ticketid, t.type AS ticket_type, e.description AS event_name, e.event_date
FROM tickets t
LEFT OUTER JOIN orders o ON t.orderid = o.orderid
JOIN events e ON t.eventid = e.eventid
WHERE t.orderid IS NULL;
```

```
SELECT e.dept, SUM(p.amtpaid) AS total_department_payroll
FROM employees e
JOIN emppayroll p ON e.empid = p.empid
GROUP BY e.dept;
```

```
SELECT e.EventID, e.Description, COUNT(t.TicketID) AS TotalTicketsSold
FROM Events e
JOIN Tickets t ON e.EventID = t.EventID
GROUP BY e.EventID, e.Description
HAVING COUNT(t.TicketID) > 5;
```

```
CREATE INDEX idx_event_date ON events(event_date);
```

```
CREATE OR REPLACE VIEW vip_customer_directory AS
SELECT name, email, phone
```

```
FROM customers
WHERE type = 'VIP';
```

```
CREATE OR REPLACE TRIGGER prevent_double_booking
BEFORE INSERT ON events
FOR EACH ROW
DECLARE
    v_count NUMBER;
BEGIN
    SELECT COUNT(*)
    INTO v_count
    FROM events
    WHERE venueid = :NEW.venueid
    AND event_date = :NEW.event_date;

    IF v_count > 0 THEN
        RAISE_APPLICATION_ERROR(-20001, 'Error: The venue is already booked for this
date.');
```

```
    END IF;
END;
/
```